

```
1 getwd()
2
3 ## Set directory
4 setwd("C:\\Users\\User\\Desktop\\IT24103771 LAB -9")
5 getwd()
6
7 ## =====
8 ## Exercise: Cookie baking time
9 ## =====
10 ## Parameters
11 mean_time <- 45
12 sd_time <- 2
13 sample_size <- 25
14
15
16 ## i. Generate random sample
17 set.seed(123)
18 sample_data <- rnorm(sample_size, mean = mean_time, sd = sd_time)
19 print(sample_data)
20
21
22 ## ii. One-tailed t-test: H0: mean = 46, H1: mean < 46
23 t_test <- t.test(sample_data, mu = 46, alternative = "less")
24 print(t_test)
25
26
```

```
> setwd("C:\\Users\\User\\Desktop\\IT24103771 LAB -9")
> getwd()
[1] "C:/Users/User/Desktop/IT24103771 LAB -9"
> mean_time <- 45
> sd_time <- 2
> sample_size <- 25
> set.seed(123)
```

```
> print(sample_data)
[1] 43.87905 44.53965 48.11742 45.14102 45.25858 48.43013 45.92183
[8] 42.46988 43.62629 44.10868 47.44816 45.71963 45.80154 45.22137
[15] 43.88832 48.57383 45.99570 41.06677 46.40271 44.05442 42.86435
[22] 44.56405 42.94799 43.54222 43.74992
> t_test <- t.test(sample_data, mu = 46, alternative = "less")
> print(t_test)
```

One Sample t-test

```
data: sample_data
t = -2.8167, df = 24, p-value = 0.004776
alternative hypothesis: true mean is less than 46
95 percent confidence interval:
 -Inf 45.58124
sample estimates:
mean of x
 44.93334
```